

Video 8.1.1

Two big theorems about determinants.

Our next goal is to understand and explain two big theorems which are a big part of the reason determinants are so useful in linear algebra.

The two theorems are closely related so we'll study them together.

First let's meet the theorems.

Big Determinant Theorem #1 ('Determinant detects invertibility').

Let A be an arbitrary $n \times n$ matrix.

Then:

A is invertible if and only if $\det A \neq 0$.

Don't forget the precise definition of " A is invertible":

It is: There exists a second matrix A^{-1} satisfying the equations

$$AA^{-1} = A^{-1}A = I_n.$$

Quick test of understanding:

Which of the following matrices are invertible?

1. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ The determinant is -2 . This is non-zero so YES the matrix is invertible

2. $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 1 & 1 & 1 \end{bmatrix}$ Two rows are equal so the determinant is 0. So NO the matrix is not invertible.

3. $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 3 & 2 \end{bmatrix}$ The determinant can be quickly calculated to be $(-1) \cdot \frac{1}{2} = -1$. This is non-zero so YES this matrix is invertible.

4. $\begin{bmatrix} 1 & 8 & 6 & 6 \\ 5 & 0 & -10 & 1 \\ -7 & 2 & 8 & 1 \end{bmatrix}$ Only square matrices can be invertible. So no need to think about determinants. NO this matrix is not invertible.

Big Determinant Theorem #2 ('Determinant respects multiplication').

Let A and B be arbitrary $n \times n$ matrices.

Then:

$$\det(AB) = \det A \det B.$$

Example:

Note that: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -1 & -3 \\ -2 & -4 \end{bmatrix} = \begin{bmatrix} -5 & -11 \\ -11 & -25 \end{bmatrix}$.

So based on this theorem we expect that:

$$\det\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\right) \det\left(\begin{bmatrix} -1 & -3 \\ -2 & -4 \end{bmatrix}\right) = \det\left(\begin{bmatrix} -5 & -11 \\ -11 & -25 \end{bmatrix}\right)$$



$(-2) \qquad (-2) \qquad (-5)(-25) - (-11)(-11) = 4.$

Goal of this lecture:

Explain why these theorems are true (by building some proofs).

Video 8.1.2

First step: The important special case of elementary matrices

We will explain these two big theorems by once again applying the powerful theory of *elementary matrices*.

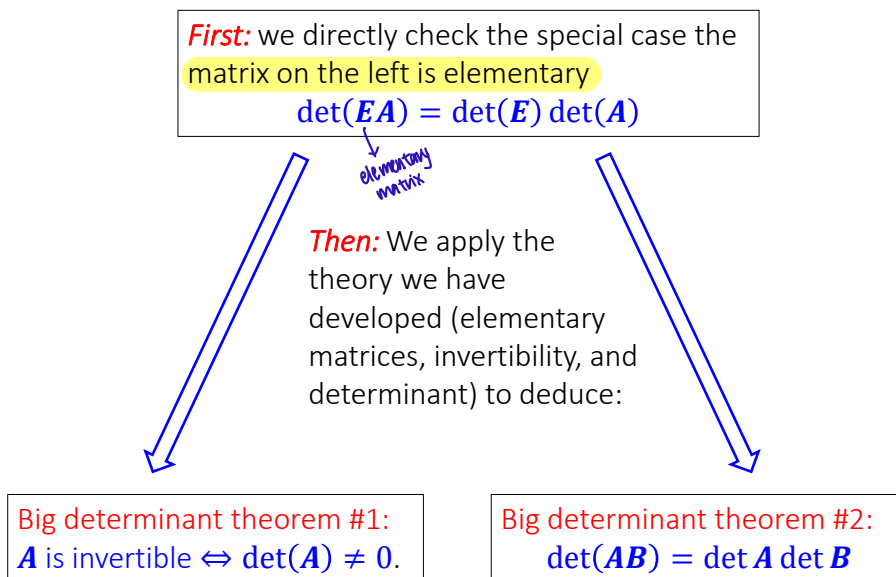
The heart of both proofs is to first verify the following important special case of **Big Determinant Theorem #2**. This is just the special case where the matrix on the left is an *elementary matrix*. We'll see that everything follows from this special case.

Lemma (A special case of Big Determinant Theorem #2.)

Let E be an arbitrary elementary $n \times n$ matrix and let A be an arbitrary $n \times n$ matrix.

Then:

$$\det(EA) = \det E \det A.$$



Let's focus on the equation we want to understand:

$$\det(\mathbf{EA}) = \det(\mathbf{E}) \det(\mathbf{A}).$$

Do you remember what this matrix $\mathbf{EA}$ is?

Answer:

Remember that that $\mathbf{EA}$ is just the matrix $\mathbf{A}'$ you get when you apply the elementary row operation corresponding to $\mathbf{E}$ to the matrix $\mathbf{A}$.

So to check this equation we'll basically need to understand: *what effect do elementary row operations have on determinants?*

$$\textcircled{1} \det \begin{pmatrix} 1 & 2 & 3 \\ -5 & 10 & -15 \\ 3 & 6 & 9 \end{pmatrix} = \det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 20 & 0 \\ 3 & 6 & 9 \end{pmatrix}$$

$$\textcircled{2} \det \begin{pmatrix} 1 & 0 & 11 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{pmatrix} = (-1) \det \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 0 & 1 \end{pmatrix}$$

$$\textcircled{3} \det \begin{pmatrix} 100 & 0 & 100 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{pmatrix} = 100 \det \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{pmatrix}$$

Video 8.1.3

Lemma (Effect of elementary row operations on determinants.)

Let A be an arbitrary $n \times n$ matrix.

The following three statements tell how the determinant of A is affected by the three types of elementary row operations.

1. If A' denotes the matrix obtained from A by an elementary row operation $R_i \rightarrow R_i + cR_j$ then

$$\det A' = \det A.$$
2. If A' denotes the matrix obtained from A by an elementary row operation $R_i \leftrightarrow R_j$ then

$$\det A' = -\det A.$$
3. If A' denotes the matrix obtained from A by an elementary row operation $R_i \rightarrow cR_i$ then

$$\det A' = c \det A.$$

Exercise:

This problem appeared on a Quiz in 2018.

Let A be a square matrix and let B be the matrix obtained from it by the following sequence of elementary row operations:

1. $R_1 \rightarrow 2R_1$
2. $R_1 \leftrightarrow R_2$
3. $R_3 \rightarrow R_3 - 2R_2$
4. $R_1 \leftrightarrow R_4$
5. $R_1 \rightarrow 2R_1$
6. $R_1 \leftrightarrow R_2$

If $\det(B) = 100$, then what is $\det(A)$?

Solution:

Let M_i denote the matrix you get after applying steps 1 through i .
 For example $B = M_6$.

$$\begin{array}{l}
 \text{B} \xrightarrow{\det(B)=100} \\
 \uparrow R_1 \leftrightarrow R_2 = (-1) \det(M_5) \\
 M_5 \\
 \uparrow R_1 \rightarrow 2R_1 = (-1)^2 2 \det(M_4) \\
 M_4 \\
 \uparrow R_1 \leftrightarrow R_4 = (-1)^2 2 \det(M_3) \\
 M_3 \\
 \uparrow R_3 \rightarrow R_3 - 2R_2 = (-1)^3 2^2 \det(M_2) \\
 M_2 \\
 \uparrow R_1 \leftrightarrow R_2 = (-1)^3 2^2 \det(M_1) \\
 M_1 \\
 \uparrow R_1 \rightarrow 2R_1 = (-1)^3 2^3 \det(A) \\
 A \\
 \therefore \det(A) = -25
 \end{array}$$

Solution:

We can start with the final matrix **B** and work back to **A**:

$$\begin{aligned}
 \det(B) &= \det(M_6) \\
 &= (-1) \det(M_5) && (M_6 \text{ results from } R_1 \leftrightarrow R_2 \text{ on } M_5) \\
 &= (-1)(2) \det(M_4) && (M_5 \text{ results from } R_1 \rightarrow 2R_1 \text{ on } M_4) \\
 &= (-1)^2(2) \det(M_3) && (M_4 \text{ results from } R_1 \leftrightarrow R_4 \text{ on } M_3) \\
 &= (-1)^2(2) \det(M_2) && (M_3 \text{ results from } R_3 \rightarrow R_3 - 2R_2 \text{ on } M_2) \\
 &= (-1)^3(2) \det(M_1) && (M_2 \text{ results from } R_1 \leftrightarrow R_2 \text{ on } M_1) \\
 &= (-1)^3(2)^2 \det(A) && (M_1 \text{ results from } R_1 \rightarrow 2R_1 \text{ on } A)
 \end{aligned}$$

Thus: $\det(B) = -4 \det(A)$. Thus: $\det(A) = -25$.

Video 8.1.4

We'll prove the 3 statements in turn simply by direct calculation using properties of determinants we understand already.

Proof for the first kind of elementary row operation:

Assume the row operation in question is: $R_i \rightarrow R_i + cR_j$.

Consider the square matrix **A** with its rows labelled as follows:

$$A = \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_i \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix}.$$

Proof for the first kind of elementary row operation (cont.):

We know that determinants are linear in rows so we calculate:

$$\det A' = \det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_i + cR_j \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix} \rightarrow \text{linear function of this row}$$

$$= \det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_i \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix} + c \det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix} \quad \begin{array}{l} * \text{linearity property of determinants} \\ \text{two identical rows} \end{array}$$

Proof for the first kind of elementary row operation (cont.):

$$= \det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_i \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix} + c \det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix}$$

↑
This matrix
is just **A**.

↑
This determinant is **zero**
because we proved earlier
that a determinant is zero
when two rows are
identical.

$$= \det A.$$

Proof for the second kind of elementary row operation. (This is just a reminder: we did this already in an earlier lecture.)

Let $\mathbf{A}$ be an arbitrary $n \times n$ matrix and let $\mathbf{A}'$ be the result of the row operation: $R_i \leftrightarrow R_j$.

$$\mathbf{A} = \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ & \vdots & \\ \leftarrow & R_i & \rightarrow \\ & \vdots & \\ \leftarrow & R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_n & \rightarrow \end{bmatrix} \quad \text{and} \quad \mathbf{A}' = \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ & \vdots & \\ \leftarrow & R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_i & \rightarrow \\ & \vdots & \\ \leftarrow & R_n & \rightarrow \end{bmatrix}.$$

We wish to prove that $\det(\mathbf{A}') = -\det(\mathbf{A})$.

We can prove this by calculating the following determinant two different ways:

$$\det \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ & \vdots & \\ \leftarrow & R_i + R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_i + R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_n & \rightarrow \end{bmatrix} = 0$$

Proof for the second kind of elementary row operation (cont.):

On the one hand we know that because two rows are equal then:

$$\det \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ & \vdots & \\ \leftarrow & R_i + R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_i + R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_n & \rightarrow \end{bmatrix} = 0.$$

On the other hand we know that because determinant is linear with respect to rows, we can expand:

$$\det \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ & \vdots & \\ \leftarrow & R_i + R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_i + R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_n & \rightarrow \end{bmatrix} = \det \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ & \vdots & \\ \leftarrow & R_i & \rightarrow \\ & \vdots & \\ \leftarrow & R_i + R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_n & \rightarrow \end{bmatrix} + \det \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ & \vdots & \\ \leftarrow & R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_i + R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_n & \rightarrow \end{bmatrix}.$$

Proof for the second kind of elementary row operation (cont.):

Then expand again to obtain:

$$\det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_i \rightarrow \\ \vdots \\ \leftarrow R_i \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix} + \det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_i \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix} + \det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_i \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix} + \det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix}.$$


= 0 because two rows are equal.


= 0 because two rows are equal.

$$\text{Thus: } 0 = \det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_i \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix} + \det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_i \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix}, \text{ as required.}$$

$$\det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_i \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix} = - \det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_j \rightarrow \\ \vdots \\ \leftarrow R_i \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix}$$

Proof for the third kind of elementary row operation:

This is just a quick application of the fact we know that determinants are linear with respect to rows.

Consider a square matrix $\mathbf{A}'$ obtained from a square matrix $\mathbf{A}$ by multiplying R_i by some constant c :

$$\mathbf{A}' = \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow cR_i \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix}.$$

We already know that:

$$\det \mathbf{A}' = \det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow cR_i \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix} = c \det \begin{bmatrix} \leftarrow R_1 \rightarrow \\ \vdots \\ \leftarrow R_i \rightarrow \\ \vdots \\ \leftarrow R_n \rightarrow \end{bmatrix} = c \det \mathbf{A}.$$

□

Video 8.1.5

Exercise (The determinants of the elementary matrices):

What are the determinants of the elementary matrices?

Case 1:

Let $\mathbf{M}$ be the elementary matrix associated to the row operation $R_i \rightarrow R_i + cR_j$ for some $c \in \mathbb{R}$.

$$\det(\mathbf{M}) = 1$$

Explanation

We know that $\mathbf{M}$ is obtained from the identity matrix $\mathbf{I}$ by the same row operation $R_i \rightarrow R_i + cR_j$. Thus:

$$\det(\mathbf{M}) = \det(\mathbf{I}) = 1.$$

Exercise (The determinants of the elementary matrices):

What are the determinants of the elementary matrices?

Case 2:

Let $\mathbf{M}$ be the elementary matrix associated to the row operation $R_i \leftrightarrow R_j$.

$$\det(\mathbf{M}) = -1$$

Explanation

We know that $\mathbf{M}$ is obtained from the identity matrix $\mathbf{I}$ by the same row operation $R_i \leftrightarrow R_j$. Thus:

$$\det(\mathbf{M}) = (-1) \det(\mathbf{I}) = (-1)(1).$$

Exercise (The determinants of the elementary matrices):

What are the determinants of the elementary matrices?

Case 3:

Let $\mathbf{M}$ be the elementary matrix associated to the row operation $R_i \rightarrow cR_i$ for some $c \in \mathbb{R}$ with $c \neq 0$.

$$\det(\mathbf{M}) = c$$

Explanation

We know that $\mathbf{M}$ is obtained from the identity matrix $\mathbf{I}$ by the same row operation $R_i \rightarrow cR_i$. Thus:

$$\det(\mathbf{M}) = (c) \det(\mathbf{I}) = (c)(1).$$

Summary with examples of determinants of elementary matrices.

The case $R_i \rightarrow R_i + R_j$. $\det \begin{pmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{pmatrix} = 1.$

The case $R_i \rightarrow R_j$. $\det \begin{pmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{pmatrix} = -1.$

The case $R_i \rightarrow cR_i$. $\det \begin{pmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2020 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{pmatrix} = 2020.$

Video 8.1.6

The special case of elementary matrices: the proof

We now know enough to prove this important fact we mentioned earlier. Recall that this is a special case of the theorem which says that **determinant is multiplicative**.

Lemma (Special case of the big determinant theorem #2.)

Let E be an arbitrary elementary $n \times n$ matrix and let A be an arbitrary $n \times n$ matrix.

Then:

$$\det(EA) = \det E \det A.$$

Lemma

Let E be an arbitrary elementary $n \times n$ matrix and let A be an arbitrary $n \times n$ matrix.

Then:

$$\det(EA) = \det E \det A.$$

Proof:

Consider the elementary row operation corresponding to the elementary matrix E . For the discussion denote the resulting of applying this on some matrix A by $R(A)$.

We will use two facts we know from the basic properties of elementary matrices. They are:

1. $EA = R(A)$.
2. $E = R(I_n)$.

Quick reminder of these two operations:

1. $\mathbf{EA} = \mathbf{R}(\mathbf{A})$.

Multiplying a matrix $\mathbf{A}$ on the left by the matrix $\mathbf{E}$ has the result of performing the corresponding elementary operation on $\mathbf{A}$.

2. $\mathbf{E} = \mathbf{R}(\mathbf{I}_n)$.

To get the elementary matrix $\mathbf{E}$ perform the row operation it corresponds to to the identity matrix.

Proof (continued):

The equation we want to show for each elementary row operation is:

$$\det(\mathbf{EA}) = \det(\mathbf{E}) \det(\mathbf{A}) \rightarrow \text{trying to prove this eqn}$$

Use fact 1 to
replace by $\mathbf{R}(\mathbf{A})$.

Use fact 2 to
replace by $\mathbf{R}(\mathbf{I}_n)$.

So we must show for all 3 elementary operations:

$$\det(\mathbf{R}(\mathbf{A})) = \det(\mathbf{R}(\mathbf{I}_n)) \det(\mathbf{A})$$

Proof (continued):

It will be easy to just check

$$\det(R(A)) = \det(R(I_n)) \det(A)$$

directly with simple equalities for all three possibilities for the elementary row operation R .

Possibility #1: The operation $R_i \rightarrow R_i + cR_j$.

We have shown this operation has **no effect on determinant**. Thus:

$$\begin{aligned} LHS &= \det R(A) \\ &= \det A \\ &= 1 \cdot \det A \\ &= \det I_n \cdot \det A \\ &= \det R(I_n) \det A = RHS. \end{aligned}$$

Possibility #2: The operation $R_i \rightarrow R_j$.

We have shown this operation multiplies determinant by -1 . Thus:

$$\begin{aligned} LHS &= \det R(A) \\ &= (-1) \det A \\ &= (-1) \cdot 1 \cdot \det A \\ &= (-1) \cdot \det I_n \cdot \det A \\ &= \det R(I_n) \det A = RHS. \end{aligned}$$

Possibility #3: The operation $R_i \rightarrow cR_i$.

We have shown this operation multiplies determinant by c . Thus:

$$\begin{aligned}
 LHS &= \det R(\mathbf{A}) \\
 &= c \det \mathbf{A} \\
 &= c \cdot 1 \cdot \det \mathbf{A} \\
 &= c \cdot \det \mathbf{I}_n \cdot \det \mathbf{A} \\
 &= \det R(\mathbf{I}_n) \det \mathbf{A} = RHS.
 \end{aligned}$$



Now that we have proved this lemma $\det(\mathbf{EA}) = \det \mathbf{E} \det \mathbf{A}$, it will be straightforward to deduce the two big determinant theorems.

Video 8.1.7

Big Determinant Theorem #1 ('Determinant detects invertibility').

Let A be an arbitrary $n \times n$ matrix.

Then:

A is invertible if and only if $\det A \neq 0$.

$\det(EA) = \det(E) \det(A)$
This fact will let us relate $\det(A)$ to $\det(\text{REF}(A))$.

Proof:

Before explaining the two cases of this proof, we will derive a useful formula for $\det A$.

Let A' be the row-reduced echelon form of A .

If we express the row reduction that turns A into A' via elementary matrices then we obtain a formula:

$$A' = E_k \dots E_1 A.$$

Proof (continued):

It is convenient to put the matrix we are studying A on the LHS of this formula:

$$A = E_1^{-1} \dots E_k^{-1} A'.$$

Because these E_i^{-1} are again elementary matrices we can now use the lemma we checked already $\det EB = \det E \det B$ to write:

$$\det A = \underbrace{\det(E_1^{-1}) \dots \det(E_k^{-1})}_{\neq 0} \det A' \quad (\star).$$

$\neq 0$ determinants of elementary matrices can never be zero. We proved it just now.

Now we can focus on the proof. We will now split the work of the proof into the two cases:

1. A is invertible $\Rightarrow \det A \neq 0$.
2. A is not invertible $\Rightarrow \det A = 0$.

Proof (continued):

(Case 1.)

What are we assuming?

We assume that A is invertible.

What do we have to show follows from this?

We want to show that $\det A \neq 0$.

REF of A is identity matrix if A is invertible.

Because A is invertible we know from the theorem on "Characterisation of invertibility" that the row reduced echelon form of A is the identity $A' = I_n$.

Thus, equation $(\star)$ becomes:

$$\det A = \det E_1^{-1} \dots \det E_k^{-1} \det I_n.$$

Every factor on the right is non-zero, so $\det A \neq 0$.

(Case 2.)

What are we assuming?

We assume A is not invertible.

What do we have to show follows from this?

Need to show that $\det A = 0$.

If A is not invertible then its row-reduced echelon form A' is not the identity matrix.

Thus, because

- A' is square
- A' is in reduced row echelon form
- A' is not the identity

** there will be one column without leading entry!
→ if that, it can only be an identity matrix*

it will have at least one row of zeros. Thus $\det A' = 0$.

There are more columns than leading entries

Thus, $(\star)$ becomes:

$$\det A = \det E_1^{-1} \dots \det E_k^{-1} \det A' = 0.$$



Video 8.1.8

Big Determinant Theorem #2 ('Determinant respects multiplication').

Let A and B be arbitrary $n \times n$ matrices.

Then:

$$\det(AB) = \det A \det B.$$

Proof:

It will again be convenient to split this into two cases.

1. A is invertible.
2. A is not invertible.

The task:
 $\det(AB) = \det(A)\det(B)$
 We'll use:
 The special case that A is elementary.
 A is invertible $\Leftrightarrow \det A \neq 0$

(Case 1: A is invertible.)

Because A is invertible we know from the theorem on "Characterisation of invertibility" that A can be expressed as a product of elementary matrices:

$$A = E_1 \dots E_k.$$

Thus:

$$\begin{aligned} \det(AB) &= \det(E_1 \dots E_k B) \\ &= \det(E_1) \det(E_2 \dots E_k B) \\ &\vdots \\ &= \underbrace{\det E_1 \dots \det E_k}_{\det(A)} \det B \quad (\leftarrow) \\ &= \underbrace{\det(E_1 \dots E_k)}_A \det B \quad (\leftarrow) \\ &= \det A \det B. \end{aligned}$$

(Steps marked $(\leftarrow)$ are some applications of the special case we proved already that $\det EB = \det E \det B$ for elementary matrices E .)

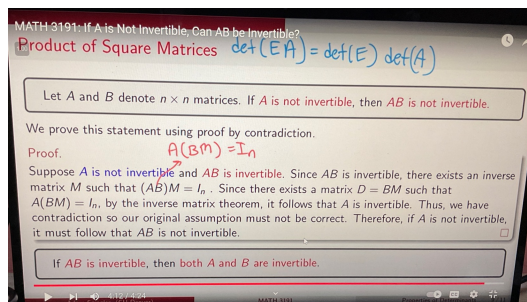
(Case 2: A is not invertible.)

First recall that we proved earlier in the course that if A is not invertible then neither is AB . (Do you remember why? Exercise!)

So in this case $\det A = \det(AB) = 0$.

Because of this the equation we must check is trivial in this case, because both sides are zero:

$$\det A \det B = \det AB.$$



Video 8.1.9

Corollary ('Determinants of inverses').

Let A be an invertible square matrix.

(So we now know that $\det A \neq 0$ and A^{-1} exists.)

Then:

$$\det A^{-1} = \frac{1}{\det A}.$$

This is a quick deduction from the theorem that says determinants are multiplicative:

$$\begin{aligned} \det A \cdot \det A^{-1} &= \det(AA^{-1}) \quad (\text{Dets. are mult.}) \\ &= \det(I_n) \\ &= 1. \end{aligned}$$

Adjoint of a Matrix

Video 8.2.1

A direct formula for the inverse matrix using determinants.

There is a way to *directly* write down the inverse of matrix using determinants.

This formula may not be as useful as it looks at first glance. Because it is based on determinants, the amount of computation you need to do to use it grows *very quickly* as the matrix becomes bigger.

In addition, it is more like a clever guess than a reflection of deeper properties of matrices. Anyway it is interesting and concrete, so we will learn it.

Definition ('The Adjoint matrix').

Let $\mathbf{A}$ be an $n \times n$ matrix.

Let C_{ij} denote the cofactor associated to the (i, j) -entry of the matrix. (Recall this is just $(-1)^{i+j}$ times the determinant of the matrix you get when you cross out row i and column j .)

Let $\mathbf{C}$ denote the matrix of cofactors (i.e. just the $n \times n$ matrix whose (i, j) -entry is C_{ij}).

The **adjoint matrix** of $\mathbf{A}$, denoted $\text{adj}(\mathbf{A})$, is defined to be the **TRANSPOSE** of the matrix of cofactors:

$$\text{adj}(\mathbf{A}) = (\mathbf{C})^T.$$

Theorem ('The Adjoint matrix is almost the inverse').

Let $\mathbf{A}$ be an $n \times n$ matrix. Let $\text{adj}(\mathbf{A})$ denote its adjoint.

Then:

$$\mathbf{A} \text{adj}(\mathbf{A}) = \text{adj}(\mathbf{A}) \mathbf{A} = (\det \mathbf{A}) \mathbf{I}_n.$$

↗ scalar multiplication
of identity matrix.

First we'll do an example.

After that we'll prove it by a direct calculation.

Example:

1. Verify the adjoint formula from the previous theorem

$$\mathbf{A} \text{adj}(\mathbf{A}) = \det(\mathbf{A}) \mathbf{I}$$

in the special case

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

2. Using this computation, write down an inverse matrix to $\mathbf{A}$.

Example:

1. Verify the adjoint formula from the previous theorem

$$\mathbf{A} \operatorname{adj}(\mathbf{A}) = \det(\mathbf{A}) \mathbf{I}$$

in the special case

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{bmatrix}. \quad \text{Matrix of cofactors} = \begin{bmatrix} 2 & (-1)^1 \cdot 0 & (-1)^{1+2} \cdot 2 \\ -1 & 1 & -2 \\ 0 & -0 & 2 \end{bmatrix}$$

2. Using this computation, write down an inverse matrix to $\mathbf{A}$.

Solution to Part 1:

The formula uses the determinant, so let's calculate that quickly:

$$\det(\mathbf{A}) = \det \left(\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{bmatrix} \right) = 2.$$

Solution to Part 1 (cont.):

Next let's calculate the adjoint matrix $\operatorname{adj}(\mathbf{A})$.

First we calculate the cofactor matrix, then we take it's transpose:

$$\begin{bmatrix} (+1) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{vmatrix} & (-1) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{vmatrix} & (+1) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{vmatrix} \\ (-1) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{vmatrix} & (+1) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{vmatrix} & (-1) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{vmatrix} \\ (+1) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{vmatrix} & (-1) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{vmatrix} & (+1) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{vmatrix} \end{bmatrix}^T$$

$$= \begin{bmatrix} 2 & 0 & 2 \\ -1 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix}^T = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 1 & 0 \\ 2 & -2 & 2 \end{bmatrix}.$$

Solution to Part 1 (continued)

We just calculated that if $\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{bmatrix}$ then

$\det(\mathbf{A}) = 2$ and

$$\text{adj}(\mathbf{A}) = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 1 & 0 \\ 2 & -2 & 2 \end{bmatrix}.$$

Now let's check the key formula $\mathbf{A} \text{adj}(\mathbf{A}) = \det(\mathbf{A}) \mathbf{I}$.

$$\mathbf{A} \text{adj}(\mathbf{A}) = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 0 \\ 0 & 1 & 0 \\ 2 & -2 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \det(\mathbf{A}) \mathbf{I}_3$$

As expected. The other formula $\text{adj}(\mathbf{A}) \mathbf{A} = \det(\mathbf{A}) \mathbf{I}$ can also be easily checked.

Solution to Part 2

Now our job is to calculate the inverse of this $\mathbf{A}$.

Well we just showed that $\mathbf{A} \text{adj}(\mathbf{A}) = \det(\mathbf{A}) \mathbf{I}$.

Because $\det(\mathbf{A}) \neq 0$ (in this case we calculated $\det(\mathbf{A}) = 2$) we can just divide by $\det(\mathbf{A})$ to deduce:

$$\underbrace{\mathbf{A} \left(\frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A}) \right)}_{\text{So this here is } \mathbf{A}^{-1}} = \mathbf{I}. \quad \rightarrow \text{can only do this if determinant is non-zero}$$

$\downarrow$
 if $\det(\mathbf{A}) \neq 0$

So in this example:

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A}) = \frac{1}{2} \begin{bmatrix} 2 & -1 & 0 \\ 0 & 1 & 0 \\ 2 & -2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1/2 & 0 \\ 0 & 1/2 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$



Question

We just saw that in the special that $\det(\mathbf{A}) \neq 0$ then we can just rearrange the equation

$$\mathbf{A} \operatorname{adj}(\mathbf{A}) = \operatorname{adj}(\mathbf{A}) \mathbf{A} = \det(\mathbf{A}) \mathbf{I} \quad (*)$$

to deduce that

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \operatorname{adj}(\mathbf{A}).$$

Why did we bother writing down Equation (*) then? Why didn't we just write down a formula for the inverse straight away?

Answer:

Because the equation (*) gives more information.

It says something non-trivial *even in the case* when $\det(\mathbf{A}) = 0$.

Video 8.2.2

Theorem ('The Adjoint matrix is almost the inverse').

Let $\mathbf{A}$ be an $n \times n$ matrix. Let $\operatorname{adj}(\mathbf{A})$ denote its adjoint. Then:

$$\mathbf{A} \operatorname{adj}(\mathbf{A}) = \operatorname{adj}(\mathbf{A}) \mathbf{A} = (\det \mathbf{A}) \mathbf{I}_n.$$

Proof:

Actually this is just going to be a direct calculation.

Before we do the calculation remind yourself of two things:

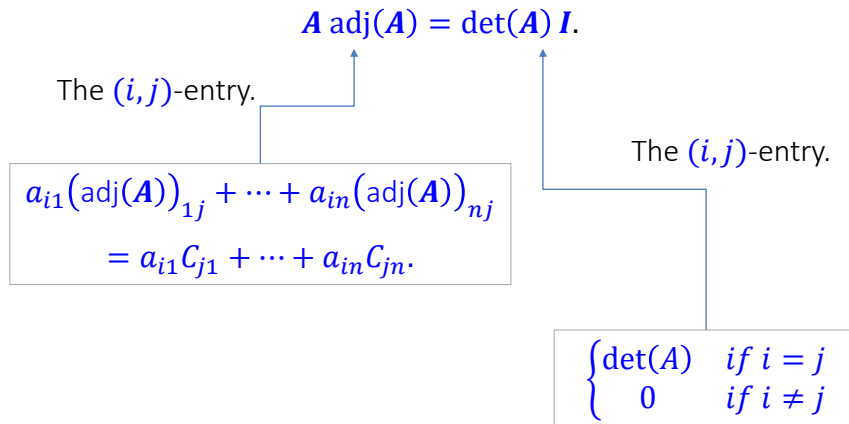
1. We just defined the (i, j) -entry of $\operatorname{adj}(\mathbf{A})$ to be the cofactor of the (j, i) -entry of $\mathbf{A}$. Namely: $(\operatorname{adj}(\mathbf{A}))_{ij} = C_{ji}$. *there's a transpose in the formula*
2. The cofactors C_{ij} are used to calculate the determinant of $\mathbf{A}$ by the usual cofactor expansion (in the next formula we can fix any row i to expand along)

$$\det \mathbf{A} = a_{i1} C_{i1} + \cdots + a_{in} C_{in}.$$

Proof (continued):

Our task is to prove the following matrix equation.

We will do that by directly checking that the (i, j) -entries of the two sides are the same for every i and j .



Proof (continued):

So we must prove:

$$a_{i1}C_{j1} + \cdots + a_{in}C_{jn} = \begin{cases} \det(\mathbf{A}) & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

The proof splits naturally into two cases at this point:

Case 1 when $i = j$ (in other words the diagonal entries).

Case 2 when $i \neq j$ (in other words the entries off the diagonal).

Proof (continued):

Case 1. So $i = j$.

We must explain why:

$$a_{i1}C_{i1} + \cdots + a_{in}C_{in} = \det(\mathbf{A})$$

Question:

Why is this
equation true?

This expression is **exactly** what
you write down when you do a
cofactor expansion for the
determinant of $\mathbf{A}$ along row i .
So obviously it equals $\det(\mathbf{A})$.

Proof (continued):

Case 2. So $i \neq j$.

We must explain why:

$$a_{i1}C_{j1} + \cdots + a_{in}C_{jn} = 0.$$

Question:

Can we recognize this expression?

*there was a replacement of a_{ij} to a_{ji}
↓
replace a_{ij} with a_{ji}*

We have to be a bit clever to identify this expression.

It looks like a cofactor expansion as normal just like in Case 1 but *there is something different here*. In a normal cofactor expansion you see terms like $a_{i1}C_{i1}$ where i appears twice but in this expression the indices are different: $a_{i1}C_{j1}$.

So in this case what does this expression correspond to?

Case 2. So $i \neq j$. We must show: $a_{i1}C_{j1} + \dots + a_{in}C_{jn} = 0$ ($\dagger$).

Explanation:

Take the matrix A . Replace row j with row i to get a new matrix A' :

$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{i1} & \dots & a_{in} \\ \vdots & \ddots & \vdots \\ a_{j1} & \dots & a_{jn} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} \rightarrow A' = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{i1} & \dots & a_{in} \\ \vdots & \ddots & \vdots \\ a_{i1} & \dots & a_{in} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix}$$

Now calculate $\det(A')$ in two different ways.

Calculation #1: A cofactor expansion along row j gives

$$\det(A') = a_{i1}C'_{j1} + \dots + a_{in}C'_{jn} = a_{i1}C_{j1} + \dots + a_{in}C_{jn}.$$

(Here we noticed the two matrices have the same cofactors along row j : $C_{jk} = C'_{jk}$.)

Calculation #2:

Note that two rows of A' coincide. Thus $\det(A') = 0$.

Observe that these two calculations give the two sides of equation ($\dagger$).

So we have proved equation ($\dagger$).



Cramer's Rule

Video 8.2.3

Consider a system of equations of the following form:

$$A \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$$

Here we assume: $\rightarrow$ same no. of eqn as variables

- A is a square $n \times n$ matrix.
- A is invertible.

Because A is invertible we know that this system has a **unique** solution. (This is one of the 4 equivalent characterisations of invertibility.)

Cramer's rule is basically a determinant trick which directly calculates the unique solution using a formula.

We'll just learn the rule here. You can explore the (easy) proof in the tutorials.

Theorem ('Cramer's Rule').

→ implies that the corresponding system has a unique solution

Let A be an invertible $n \times n$ matrix.

Let $(b_1, \dots, b_n) \in \mathbb{R}^n$.

Recall that because A is invertible the matrix equation

$$A \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$$

has a unique solution.

Let A_j denote obtained by replacing column j of A with the column

vector $\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$.

Then: for all $1 \leq i \leq n$ we have: $x_i = \frac{\det(A_i)}{\det(A)}$ is the x_i component of the unique solution.

Exercise:

Use Cramer's rule to calculate the unique solution to the system of linear equations:

$$\begin{cases} 2x + y + z &= 3 \\ x - y - z &= 0 \\ x + 2y + z &= 0 \end{cases}$$

Exercise:

Use Cramer's rule to calculate the unique solution to the system of linear equations:

$$\begin{cases} 2x + y + z = 3 \\ x - y - z = 0 \\ x + 2y + z = 0 \end{cases}$$

Solution:

The first thing to do is to rewrite this as an equivalent matrix equation:

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}.$$

Solution (cont.)

We'll need the determinant of the matrix on the LHS:

$$\det \begin{pmatrix} 2 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & 2 & 1 \end{pmatrix} = 2(-1 + 2) - 1(1 + 1) + 1(2 + 1) = 3.$$

Note that this determinant is $\neq 0$ so the system does indeed have a unique solution.

Cramer's rule tells us the components of the unique solution are:

$$x_1 = \frac{\det \begin{pmatrix} 3 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 2 & 1 \end{pmatrix}}{\det \begin{pmatrix} 2 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & 2 & 1 \end{pmatrix}} = \frac{3(-1 + 2)}{3} = 1.$$

The matrix on the top line is what you get after you replace column 1 of the LHS matrix with the RHS vector.

The matrix on the bottom line is just the matrix on the LHS of the original equation.

Solution (cont.)

$$x_1 = \frac{\det \begin{pmatrix} \textcolor{red}{3} & 1 & 1 \\ \textcolor{red}{0} & -1 & -1 \\ \textcolor{red}{0} & 2 & 1 \end{pmatrix}}{\det \begin{pmatrix} 2 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & 2 & 1 \end{pmatrix}} = \frac{3(-1+2)}{3} = 1.$$

$$x_2 = \frac{\det \begin{pmatrix} 2 & \textcolor{red}{3} & 1 \\ 1 & \textcolor{red}{0} & -1 \\ 1 & \textcolor{red}{0} & 1 \end{pmatrix}}{\det \begin{pmatrix} 2 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & 2 & 1 \end{pmatrix}} = \frac{-3(1+1)}{3} = -2.$$

$$x_3 = \frac{\det \begin{pmatrix} 2 & 1 & \textcolor{red}{3} \\ 1 & -1 & \textcolor{red}{0} \\ 1 & 2 & \textcolor{red}{0} \end{pmatrix}}{\det \begin{pmatrix} 2 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & 2 & 1 \end{pmatrix}} = \frac{3(2-(-1))}{3} = 3.$$

Solution (cont.)

We just used Cramer's rule to calculate the solution

$$\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}.$$

Let's check our answer:

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}.$$

As expected.

